

Waste Collection Optimization Using Machine Learning

SDG 11 – Sustainable Cities and Communities

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1. Introduction

Rapid urbanization has significantly increased the amount of municipal solid waste generated every day. Traditional waste collection systems follow fixed schedules regardless of whether waste bins are full or empty. This results in inefficient fuel consumption, increased operational costs, overflowing bins, and environmental pollution.

Machine Learning (ML) can optimize waste collection by predicting when waste bins will become full and recommending the most efficient collection routes. This approach supports **United Nations Sustainable Development Goal (SDG) 11 – Sustainable Cities and Communities**, which focuses on making cities cleaner, safer, more resilient, and environmentally sustainable.

2. Problem Understanding

Problem Statement

Many municipalities follow fixed waste collection schedules without considering the actual fill level of waste bins. Consequently:

- Trucks often collect half-empty bins.
- Some bins overflow before the scheduled collection.
- Fuel consumption increases.
- Collection costs become higher.
- Citizens experience poor sanitation.
- Carbon emissions increase due to unnecessary vehicle movement.

The objective is to design a Machine Learning system that predicts bin fill levels and optimizes collection routes to reduce cost, save fuel, and improve city cleanliness.

3. Objectives

The proposed system aims to:

- Predict the fill level of waste bins.
- Identify bins requiring immediate collection.
- Reduce unnecessary waste collection trips.
- Optimize vehicle routes.
- Reduce operational costs.
- Improve urban sanitation.
- Support sustainable city development under SDG 11.

4. Dataset Collection

The dataset can be collected from smart IoT-enabled waste bins or municipal records.

Sample Dataset

Bin ID	Area	Waste Level (%)	Temperature	Humidity	Last Collection (Days)	Traffic Level	Collection Required
1	Residential	92	34	65	3	High	Yes
2	Commercial	55	32	60	2	Medium	No
3	Industrial	88	36	70	4	High	Yes
4	School	40	31	55	1	Low	No
5	Hospital	97	35	72	5	High	Yes

Target Variable:

Collection Required

Yes / No

5. Data Preparation

Before training the machine learning model, the dataset must be preprocessed.

Step 1: Handle Missing Values

Missing values are replaced using:

- Mean
- Median
- Mode

depending on the attribute type.

Step 2: Remove Duplicate Records

Duplicate entries are removed to prevent biased predictions.

Step 3: Encode Categorical Variables

Categorical values are converted into numerical form.

Example:

Area

Residential → 0

Commercial → 1

Industrial → 2

School → 3

Hospital → 4

Collection Required

No → 0

Yes → 1

Step 4: Feature Scaling

Numerical features such as waste level and temperature are normalized to improve model performance.

Methods:

- Min-Max Scaling
- Standard Scaling

Step 5: Train-Test Split

The dataset is divided into:

Training Data = 80%

Testing Data = 20%

6. Data Analysis Using NumPy and Pandas

Python libraries used:

- NumPy
- Pandas

Sample analysis includes:

- Average waste level
- Maximum waste level
- Minimum waste level
- Number of bins requiring collection
- Average traffic conditions
- Area-wise waste generation
- Correlation between waste level and collection requirement

Example:

Average Waste Level

82%

Highest Waste Level

97%

Bins needing collection

3 out of 5

This analysis helps understand waste generation patterns.

7. Machine Learning Model

Since the target variable is binary (Yes/No), classification algorithms are suitable.

Algorithms considered:

Decision Tree

Advantages:

- Easy to understand
- Fast prediction
- Handles categorical data

Random Forest

Advantages:

- High accuracy
- Less overfitting
- Handles large datasets
- Robust performance

Logistic Regression

Suitable for binary classification.

Simple and interpretable.

Selected Algorithm

Random Forest Classifier

Reason:

- High prediction accuracy
- Works well on real-world datasets
- Reduces overfitting
- Handles noisy data
- Identifies important features

8. Model Training

Training Process:

1. Load dataset.
2. Perform preprocessing.
3. Select important features.
4. Split training and testing data.
5. Train Random Forest classifier.
6. Test model performance.
7. Predict bins requiring collection.

9. Model Evaluation

Performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve

Suppose the trained model produces:

- Accuracy = 94%
- Precision = 93%
- Recall = 92%
- F1 Score = 92.5%

These values indicate good prediction performance.

10. Visualization

The following graphs help analyze waste collection.

Graph 1

Bar Chart

Area vs Average Waste Level

Purpose:

Identify which areas generate the most waste.

Graph 2

Pie Chart

Collection Required

Yes vs No

Purpose:

Shows percentage of bins requiring collection.

Graph 3

Line Chart

Waste Level over Time

Purpose:

Monitor waste accumulation trends.

Graph 4

Heat Map

Feature Correlation

Purpose:

Shows relationship between waste level, traffic, temperature, and collection requirement.

11. Proposed System Architecture

Waste Bin Sensors (IoT)

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Data Collection

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Data Preprocessing

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Feature Selection

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Random Forest Model

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Fill Level Prediction

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Route Optimization

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Waste Collection Truck

12. Innovation

The proposed system introduces several innovative features:

- IoT-enabled smart waste bins
- Real-time waste monitoring
- AI-based waste prediction
- Dynamic route optimization
- Fuel-efficient waste collection

- Mobile dashboard for municipality staff
- GPS tracking of collection vehicles

Future enhancements include integrating drones for monitoring illegal dumping and using reinforcement learning for adaptive route optimization.

13. SDG 11 Mapping

SDG 11: Sustainable Cities and Communities

The proposed machine learning system contributes to SDG 11 by:

- Improving urban cleanliness.
- Reducing overflowing waste bins.
- Lowering greenhouse gas emissions.
- Optimizing transportation routes.
- Reducing fuel consumption.
- Enhancing public health.
- Supporting smart city initiatives.
- Promoting sustainable waste management.

Thus, the project directly supports the development of cleaner, healthier, and more sustainable urban environments.

14. Advantages

- Reduces waste collection cost.
- Saves fuel.
- Prevents overflowing bins.
- Improves public hygiene.
- Reduces carbon emissions.
- Supports smart city infrastructure.
- Better resource utilization.
- Improves operational efficiency.

15. Limitations

- Requires IoT infrastructure.
- Initial installation cost is high.
- Requires internet connectivity.
- Sensors require periodic maintenance.
- Large datasets are needed for better accuracy.

16. Future Scope

Future improvements include:

- Deep Learning for better prediction accuracy.
- Drone-assisted waste monitoring.
- Reinforcement Learning for dynamic route planning.
- Mobile applications for citizen reporting.
- AI-powered waste segregation.
- Integration with GIS and Google Maps for intelligent navigation.

17. Conclusion

Machine Learning provides an intelligent and efficient solution for optimizing waste collection in smart cities. By predicting waste bin fill levels and recommending optimized collection routes, municipalities can significantly reduce fuel consumption, operational costs, and environmental pollution. The proposed Random Forest-based system ensures accurate predictions while improving sanitation and resource management. This project strongly aligns with United Nations SDG 11 – Sustainable Cities and Communities, contributing to cleaner, greener, and more sustainable urban environments.